

GAY SHIN LEE

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ABOUT ME

Adaptable and driven Computer Science student at SUTD with a strong foundation in software development, human-centered design, and artificial intelligence. Experienced in full stack development, and design thinking.

EDUCATION

Singapore University of Technology and Design (SUTD)

- Bachelor of Engineering (Computer Science Design)
- September 2022 - present

CERTIFICATIONS

AWS Certified Cloud Practitioner, Amazon Web Services

Jul 2025

Google UX Design Professional Certificate, Coursera

Sep 2024

TECHNICAL SKILLS

Programming languages : Python, Java, SQL, HTML/CSS/JS

Development Tools: VSCode, PyCharm, IntelliJ, Jupyter, Git

UI/UX Tools: Figma, Sketch, Framer, InVision Studio, Lovable, Cursor

Languages: Fluent in English and Chinese language (Written & Spoken)

WORK EXPERIENCES

HTX (Home Team Science & Technology Agency)

Singapore

UX Researcher (Lead)

Sept 2025 - Present

- **Led** mixed-methods user research (N=15) with SPF frontline officers to drive the design of an offline-first Hybrid RAG system, directly resolving critical information bottlenecks and operational delays in connectivity blackspots.
- **Translated** raw field data into actionable product requirements using Affinity Mapping, and utilized the MoSCoW method to align developers and HTX stakeholders on a lean, offline-first MVP.

The National Kidney Foundation (NKF)

Singapore

UX Researcher & UI/UX Designer (Lead)

Sept - Dec 2025

- **Led** mixed-methods research (N=61) to develop a sustainable WebAR mini-game to eliminate paper brochure waste while gamifying public health education at physical exhibits.
- **Authored and conducted** a moderated usability study grounded in ISO 9241-11 and NN/g heuristics

TechCreate Group

Singapore

Full Stack Software Engineer Intern

May - Dec 2025

- Real Time Engine (FAST Instant Payments) – Heong Leong Finance
 - **Delivered** a custom RTE for FAST instant payments (ISO 20022), achieving a consistent 10-minute turnaround per file and saving 300+ man-minutes per month.
 - **Engineered** an automated end-to-end test data generation to accelerate payment system QA workflows.
 - **Built** a custom file-format conversion pipeline for ISO 20022 messages, eliminating manual preparation errors and standardizing test inputs.
 - **Designed and executed** dynamic Postman test suites across 20+ critical payment APIs, significantly raising overall test coverage and system reliability.
- Soundbox Device Activation – ABA Bank
 - **Productionized** a Python activation script that automates merchant authentication, QR issuance, and deterministic config packaging—cutting per-device activation time from ~2 hours to <15 minutes (~88% reduction).
 - **Authored** a comprehensive Functional Specification Document (FSD), translating business requirements into system behaviors, payment flows, error handling, and ISO mappings.
 - Reduced onboarding variability and increased operational throughput for device activations.

Regional Rewards Systems — DBS Bank

- **Mapped** end-to-end cross-regional user journeys and presented them to align product, engineering, and business stakeholders across regions.
- **Clarified** feature scope and dependencies to accelerate delivery and reduce rework during rollout.

DBS

Singapore

Quality Assurance Intern

Oct 2024 - Jan 2025

- QA testing for DRA (Dynamic Risk Assessment) team using Selenium Cucumber framework with Java for automation test scripts, and SQL for manual testing, which helped reduce test code redundancy.
 - Provided mock data based on rule logic and QA descriptions using Pandas.
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PROJECTS

50.006 User Interface Design and Implementation - FlowTutor

Sept - Dec 2025

- **Led** mixed method research and synthesised 2 user personas, 3 scenario archetypes (Easy/Hard/Edge)
- **Designed** user flows and high-fidelity prototypes to solve complex layout constraints, seamlessly integrating a video player, AI chatbot, and note-taking panel within a single desktop viewport.
- **Iterated** on the design after the group project ended.

Self-initiated

UX Design Project – Reimagining CaringBridge

Sep 2024

- **Redesigned** the user experience for CaringBridge, a non-profit health communication platform, utilizing empathy mapping and journey analysis to deliver a highly accessible interface for families managing medical crises.
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CO-CURRICULAR ACTIVITIES

SUTD

Singapore

Chamber Ensemble

Sept 2022 - Dec 2024

Vice President

- Worked with my team to organise for multiple events including SUTD Open House, Lunchtime performances, Mini Showcase etc, and handled various admin matters including the club's finances.
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ADDITIONAL INFORMATION

SUTD FACT Programme (DES x UGM)

Yogyakarta, Indonesia

Jan 2024

- Collaborated with the student in Gadjah Mada University (UGM) to plan and install a solar panel in Turban Village for the village to have power to pump water from the well.

Yonsei International Summer School

Seoul, South Korea

Korean Language

July 2023

- Acquired a foundational understanding of Korean grammar and vocabulary.